

Notes: Campello (JFE, 2006)

Debt Financing: Does It Boost or Hurt Firm Performance in Product Markets?

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1 Big picture

- **Question.** Does debt help firms compete in product markets, or does it hurt them?
- **Setting.** The paper combines a simple financing-and-product-market model with firm-level evidence from a panel of U.S. firms in well-defined four-digit SIC industries from 1971 to 2000.
- **Core challenge.** Earlier theory and evidence looked contradictory. Some papers argued debt makes firms aggressive; others found that highly levered firms lose ground. Campello's point is that both patterns can be true in the same data.
- **Main conceptual contribution.** Debt should not be modeled as having one linear effect on competitive performance. Instead, the debt–performance relationship is **nonmonotonic**: moderate debt can help, but excessive debt can hurt.
- **Theory mechanism.** In a setting with contract renegotiation and asymmetric information, debt-like finance can let a firm credibly threaten product-market overinvestment and push rivals to accommodate. But that logic breaks down once debt becomes too large because extreme inefficiencies are renegotiated away and/or trigger negative product-market consequences.
- **Empirical design.** The paper measures both leverage and performance *relative to industry-year rivals*. It studies whether a firm's **relative sales growth** responds to its **relative leverage**. To deal with debt endogeneity, it instruments leverage with **asset tangibility**, interpreted as creditors' liquidation value of the firm's assets.
- **Main baseline result.** Firms with moderately high leverage relative to rivals gain market share. Firms with very high leverage relative to rivals lose market share.
- **Economic magnitudes.** In the baseline GMM estimates, a one-standard-deviation increase in leverage is associated with roughly:
 - –4.1% relative sales underperformance in the **VeryLowLev** range,
 - +0.7% in the **LowLev** range,
 - +2.1% in the **HighLev** range,
 - and –16.4% in the **VeryHighLev** range.
- **Key heterogeneity result.** The nonlinearity is much stronger in concentrated industries.
- **Key leadership result.** In concentrated industries, ordinary firms with moderately high leverage gain sales, but market leaders with similar leverage do not. Low-leverage leaders, by contrast, perform better when leverage rises.
- **Main credibility result.** The result survives alternative lag structures, alternative leverage definitions, restrictions on industry-year skewness, larger-industry subsamples, firm fixed effects, and GMM specifications that also instrument investment and selling expenditures.
- **Economic takeaway.** Debt is neither uniformly strategic nor uniformly harmful. Its competitive effect depends on *how much debt the firm already carries relative to rivals*, on *market structure*, and on *the firm's market position*.

2 Data and measurement (key objects)

2.1 Industry setting and sample construction

- The data come from COMPUSTAT Primary/Supplementary/Tertiary, Full Coverage, and Research annual tapes.
- Sample period: **1971–2000**.
- Excluded firms: financial institutions, government enterprises, and not-for-profit organizations.
- The paper keeps firm-years with valid data on total assets, long-term debt, and sales.
- It removes observations with sales growth or asset growth above 200% in a single year.
- All accounting data are CPI-deflated.
- Product markets are identified at the **four-digit SIC** level following Clarke (1989). SIC codes ending in “0” or “9” are avoided when they do not describe economically meaningful markets.
- After the industry-screening step, 168 four-digit SIC industries remain in the broad sample.
- The final regression sample requires at least **10 firms per industry-year**; this yields **115 industries** and **3,201 industry-years**.

Interpretation. The paper is designed to compare firms against close rivals inside well-defined product markets rather than against the whole economy.

2.2 Performance measure: relative sales growth

The paper’s core product-market outcome is annual firm sales growth:

$$\text{Sales Growth}_{i,t} = \frac{\text{Sales}_{i,t} - \text{Sales}_{i,t-1}}{\text{Sales}_{i,t-1}}. \quad (1)$$

Rather than using raw sales growth, Campello subtracts the **industry-year mean** from each firm’s value.

Interpretation. This turns the dependent variable into a **relative-to-rival market-share style measure**. A positive value means the firm is growing faster than the average rival in that industry-year.

2.3 Leverage measure

The main debt variable is **long-term book leverage**:

$$\text{Leverage}_{i,t} = \frac{\text{Long-term debt}_{i,t}}{\text{Total assets}_{i,t}}. \quad (2)$$

Key choices:

- book values are used instead of market leverage;

- leverage is restricted to the $[0, 1]$ range;
- firms with negative book equity are dropped.

Interpretation. Long-term book leverage is less mechanically linked to near-term stock-market valuation and therefore less exposed to immediate reverse-causality concerns than market leverage.

2.4 Controls

The baseline empirical model controls for the main firm characteristics that could jointly affect financing and performance:

- **Size:** $\log(\text{assets})$,
- **Profitability:** operating earnings plus depreciation over assets,
- **Investment:** capital expenditures over assets,
- **SellExpenses:** advertising plus selling expenses over sales.

The model uses lags of profitability, investment, and sell-expenses.

Interpretation. The paper tries to isolate the product-market role of debt from the effects of firm scale, operating strength, investment policy, and past selling effort.

2.5 Tangibility as the debt instrument

The paper's instrument is **asset tangibility**, a proxy for how much creditors can recover in liquidation:

$$\text{Tangibility} = \frac{\text{Cash} + 0.715 \text{Receivables} + 0.547 \text{Inventory} + 0.535 \text{Fixed Capital}}{\text{Total assets}}. \quad (3)$$

The weights come from Berger et al. (1996), who estimate resale values of asset categories in liquidation.

Interpretation. Tangibility is meant to capture a firm's collateral value. If lenders can recover more from the asset base, they are willing to finance more leverage.

2.6 Relative-to-rival normalization

Every main variable is adjusted by subtracting its **industry-year mean**. In addition, leverage is standardized by the **industry-year standard deviation**:

$$z\text{-Leverage}_{i,t} = \frac{\text{Leverage}_{i,t} - \overline{\text{Leverage}_{j,t}}}{s(\text{Leverage})_{j,t}}. \quad (4)$$

Interpretation. A 5% debt deviation does not mean the same thing in a low-leverage industry and a high-leverage industry. The z-score identifies whether a firm's leverage is truly unusual *relative to its rivals*.

2.7 Concentration and leadership variables

In the later part of the paper, the author studies whether the main effect is stronger in some market environments.

- **Market concentration:** measured by the Herfindahl-Hirschman index (HHI) from the 1982, 1987, and 1992 U.S. Economic Censuses.
- **Concentrated industries:** $HHI \geq 1800$.
- **Competitive industries:** $HHI \leq 1000$.
- **Leader:** a firm whose beginning-of-period sales account for more than 15% of industry-year sales.

Interpretation. The idea is that strategic debt should matter more when rivals are few and when a firm's own actions influence the market more strongly.

2.8 Descriptive facts

From Table 1:

- mean assets: \$528.19 million,
- mean sales growth: 6.09%,
- mean leverage: 20.29%,
- mean tangibility: 0.542,
- total observations: about 44,556 firm-years before the tangibility requirement binds.

From Table 3:

- “very highly levered” firms in the most extreme relative-leverage range are not simply near-bankrupt outliers;
- their average raw leverage is below 29%, but their leverage is more than 3 standard deviations above the industry-year mean;
- the most extreme positive and negative leverage categories still contain many firms and observations.

Interpretation. The right tail of relative leverage is economically meaningful, not just a tiny set of distressed firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism: product-market investment and debt contracts

Campello's theory has two firms, A and B , that compete in a product market. Firm j 's payoff is

$$P_j = R(I_j, I_k) + y_j^i(I - I_j), \quad j, k \in \{A, B\}, \quad (5)$$

where:

- I_j is product-market investment,
- $R(I_j, I_k)$ is revenue from market competition,
- I is internal funds,
- y_j^i is the opportunity cost of product-market investment, with high and low states.

In the frictionless benchmark, firm A 's optimal investment solves

$$R_{I_A}(I_A^*, I_B) = y_A^i. \quad (6)$$

Interpretation. Without financing frictions, a threat to overinvest is not credible: once the time comes to act, the firm just chooses the privately optimal level.

3.2 Credit-market side

Lenders provide debt-like financing backed by liquidation value. If lenders seize the firm's assets, a fraction $1 - \ell$ of the asset value T is recoverable. The lender's profit is

$$\pi = L - rI, \quad (7)$$

where I is the amount lent, L is promised repayment, and r is the financing rate.

The model assumes:

- operating profit is observable but not verifiable,
- liquidation value is verifiable,
- lenders are less informed than the borrower about the opportunity cost state y_A ,
- contracts can be renegotiated ex post.

Interpretation. Debt matters because it is collateral-backed and because uninformed lenders cannot fully write contracts on product-market fundamentals.

3.3 Incentive compatibility and lender participation

The optimal contract must satisfy incentive compatibility for the borrower and participation for the lender. Campello writes these as:

$$y_A^k \in \arg \max R(I_A^j, I_B) - L_A^j + y_A^i(I - (1 - a)I_A^j), \quad (8)$$

and

$$(1 - r) \left[m(L_A^h - raI_A^h) + (1 - m)(L_A^l - raI_A^l) \right] \geq Z, \quad (9)$$

where a is the fraction of total investment financed externally, m is the probability of the high-opportunity-cost state, and Z is the value of gains from renegotiation.

Interpretation. The contract must give the borrower incentives to reveal her type truthfully while still making the lender willing to participate.

3.4 Three main theoretical propositions

Proposition 1. If the lender fully observes the firm's opportunity cost y_A , then no renegotiation-proof contract can support overinvestment.

Interpretation. "Informed" finance, like insider equity, cannot credibly commit the firm to an inefficiently aggressive product-market strategy because inefficiencies get renegotiated away.

Proposition 2. Under asymmetric information, there exists an optimal renegotiation-proof contract such that:

- investment is efficient if rivals accommodate or if the firm's opportunity cost is high,
- but the firm overinvests when the opportunity cost is low and the rival does not accommodate.

Interpretation. Debt can let the firm threaten rivals with aggressive conduct and thereby gain a strategic edge.

Proposition 3. The overinvestment effect has an upper bound. The low-state investment level I_A^l cannot exceed a threshold F that satisfies

$$L_A^h(F) + y_A^l - \frac{m}{1 - m}(y_A^h - y_A^l)(1 - a)F - R(F, I_B) > 0. \quad (10)$$

Interpretation. This is the formal source of the paper's nonlinearity: external finance is useful up to a point, but too much debt cannot support the strategic commitment story.

3.5 The leverage threshold

For debt to be a useful strategic device, the externally financed share a must satisfy $0 < a < \bar{a}$, where

$$\bar{a} = \frac{\frac{m}{1 - m} \Delta y_A}{r_A^h + \frac{m}{1 - m} \Delta y_A - y_A^l}, \quad \Delta y_A \equiv y_A^h - y_A^l. \quad (11)$$

Interpretation. Moderate leverage can support credible overinvestment; excessive leverage cannot.

3.6 Baseline empirical specification

The core panel regression is

$$\begin{aligned} \text{SalesGrowth}_{i,t} = & \alpha + \beta \text{Size}_{i,t} + \sum_{k=1}^2 f_k \text{Profitability}_{i,t-k} + \sum_{k=1}^2 \lambda_k \text{Investment}_{i,t-k} \\ & + \sum_{k=1}^2 g_k \text{SellExpenses}_{i,t-k} + d \text{Leverage}_{i,t-2 \text{ (or } t-3)} + \varepsilon_{i,t}. \end{aligned} \quad (12)$$

The dependent variable and all regressors are measured relative to the industry-year mean.

Interpretation. The coefficient on leverage asks: when a firm is more levered than its typical rival, does it subsequently gain or lose sales relative to those rivals?

3.7 Spline specification for nonlinearity

To allow leverage to have different marginal effects at different levels, the paper replaces the linear leverage term with a spline:

$$z\text{-Leverage}_1 = \min[z\text{-Leverage}, \ell_1], \quad (13)$$

$$z\text{-Leverage}_k = \max(\min[z\text{-Leverage}, \ell_k], \ell_{k-1}) - \ell_{k-1}, \quad k = 2, \dots, n. \quad (14)$$

The baseline four-segment specification divides leverage into:

- **VeryLowLev**: below -3σ ,
- **LowLev**: from -3σ to 0,
- **HighLev**: from 0 to $+3\sigma$,
- **VeryHighLev**: above $+3\sigma$.

Interpretation. This is the main empirical way the paper lets the data decide whether the debt effect changes sign across leverage regions.

3.8 Concentration and leadership extension

To study strategic interactions more directly, Campello augments the spline with interactions with **Leader** and estimates the model separately for concentrated and competitive industries.

Interpretation. The question becomes not just whether leverage matters, but *for whom* and *in what market environment* it matters.

3.9 Estimation choice and standard errors

The paper reports:

- instrumental-variables (IV) estimates using tangibility as the main leverage instrument,
- two-step efficient GMM estimates that also instrument investment and selling expenses,

- firm-fixed-effect specifications as an additional robustness check.

Standard errors are heteroskedasticity-robust and clustered at the firm level.

Interpretation. The econometrics are built to address debt endogeneity, possible joint endogeneity of other controls, and unobserved firm heterogeneity.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The baseline leverage effect is identified from:

- within-industry-year differences across firms,
- over-time changes in a firm's leverage relative to rivals,
- and the part of leverage predicted by tangibility.

Interpretation. The paper is not asking whether high-debt firms differ from low-debt firms in the whole economy. It asks whether firms that are unusually levered *within their own product market and year* subsequently perform differently.

4.2 Why reverse causality is weaker in the relative-to-rival design

A classic problem is that firms may change leverage because they anticipate future product-market outcomes. Campello argues the relative-to-rival setup helps because:

- unobservable industry shocks are differenced out each year,
- rivals' financing and product-market policies are outside the firm's control set,
- leverage levels, unlike event-style recapitalizations, reflect cumulative financing choices rather than only one contemporaneous shock.

Interpretation. Measuring everything relative to rivals removes much of the variation that could make sales growth and leverage move together for purely industry-wide reasons.

4.3 Why tangibility is a plausible instrument

The exclusion restriction is:

Conditional on controls and industry-year effects, a firm's asset tangibility should affect relative sales growth mainly through the firm's financing capacity.

Why the paper finds this plausible:

- lenders directly care about the liquidation value of hard assets;
- tangibility strongly predicts leverage;

- the instrument is motivated by the same contracting friction as the theory.

The first-stage relation is economically meaningful: the paper reports a leverage-on-tangibility slope of about 1.283 with a t -statistic of 4.22.

Interpretation. This is one of the paper’s main strengths. The instrument is not borrowed from a distant setting; it comes directly from the contract-theory logic used in the model.

4.4 How the paper deals with skewness and small-industry problems

Because leverage is z-scored within industry-year, the standard deviation itself matters. The author therefore checks:

- industry-years with at least 40 firms,
- industry-years whose leverage skewness lies in $[-1, 1]$.

Interpretation. The nonlinearity is not a mechanical artifact of noisy standard-deviation estimates in thin or highly skewed cross sections.

4.5 Additional robustness checks

The paper also shows the main pattern survives:

- using leverage lagged three years instead of two,
- using total debt instead of long-term debt,
- adding firm fixed effects,
- instrumenting other controls with GMM.

Table 7 reports Hansen J -test p -values between 0.20 and 0.88, and first-stage F -statistics comfortably above conventional weak-instrument thresholds.

Interpretation. The sign pattern—mildly positive in the middle, strongly negative in the far right tail—does not depend on one fragile specification.

4.6 Local distress versus strategic debt

A natural concern is that the **VeryHighLev** region may just contain distressed firms. Table 5 helps counter that concern:

- median profitability is positive even in the **VeryHighLev** range,
- many firms sit in that range,
- average raw leverage there is below 29%, so these are not absurdly levered firms in absolute terms.

Interpretation. The negative coefficient at very high relative leverage looks like a genuine strategic underperformance result, not just a disguised distress story.

4.7 What the paper can and cannot distinguish

- The paper is strong on **whether** debt has a nonlinear relation with product-market performance.
- It is weaker on pinning down the exact **product-market instrument** firms use—price, advertising, output, capacity, R&D, or distribution.
- It does not formally model every negative channel that may dominate at high leverage; some of those come from earlier papers on predation, customer concerns, or debt overhang.

Interpretation. The paper is a reduced-form statement about competitive outcomes. It says a lot about *what leverage does* but less about *exactly how firms operationalize it*.

4.8 Relation to the literature

The paper’s empirical contribution is to reconcile earlier findings:

- Opler and Titman-style evidence that highly levered firms underperform,
- Campello and Fluck-style evidence that relatively highly levered firms may outperform.

It reconciles them by showing that one finding applies in the far right tail, while the other applies in the moderate-high leverage range.

Interpretation. The earlier literature was not necessarily contradictory. It may have been looking at different parts of the same nonlinear curve.

5 Tables and figures

5.1 Table 1: descriptive statistics for the overall sample

Read:

- Mean leverage is 20.29%, with a median of 17.10%.
- Mean sales growth is 6.09%, though the distribution is wide.
- Mean tangibility is 0.542.
- The raw sample contains about 44,556 firm-years.

Economic message. The paper works with a broad cross-industry panel and a leverage distribution that is not dominated by extreme observations.

5.2 Table 2: why earlier papers appeared to disagree

Read:

- When leverage is measured as an “absolutely high leverage” dummy (top three deciles), the coefficient is negative.

Table 1

Descriptive statistics for the overall sample

This table reports summary statistics for the main variables used in the regression estimations (before industry-year adjustments). The data are collected annually from COMPUSTAT. Firm *Size* is total assets (COMPUSTAT item #6). *Sales Growth* is the annual gross sales growth at time t , given by $(Sales_t - Sales_{t-1})/Sales_{t-1}$, where *Sales* is item #12. *Leverage* is the ratio of long-term debt (item #9) to total assets. *Profitability* is operating earnings plus depreciation (item #18 + item #14) over assets. *Investment* is capital expenditures (item #172) over assets. *SellExpenses* is the ratio of advertising (item #45) and selling expenses (item #189) to total sales. *Tangibility* is computed as a weighted sum of cash holdings, accounts receivables, inventories, and net fixed capital, divided by total assets ($=$ item #1 $\times$ 0.715 + item #2 $\times$ 0.547 + item #3 $\times$ 0.535 + item #8)/item #6). The sample period is 1971–2000. Included firms are from industries selected at the four-digit SIC level following Clarke (1989). Only observations from industry-years containing at least ten firms are used.

	Mean	Median	Std. Dev.	Pct. 25	Pct. 75	N. Obs.
<i>Size</i> (\$ Millions)	528.19	34.77	2695.33	7.20	178.84	44,556
<i>Sales Growth</i>	0.0609	0.0437	0.2675	−0.0525	0.1790	44,556
<i>Leverage</i>	0.2029	0.1710	0.1774	0.0538	0.3016	44,556
<i>Profitability</i>	0.0215	0.0853	0.4561	0.0299	0.1287	44,133
<i>Investment</i>	0.0840	0.0601	0.0829	0.0306	0.1097	44,001
<i>SellExpenses</i>	0.2145	0.1597	0.3719	0.0511	0.3579	43,514
<i>Tangibility</i>	0.5421	0.5498	0.1127	0.4978	0.5975	42,994

z-score) it within each industry-year.¹⁵ In other words, I take the approach that it is not the *absolute size* of the difference from the industry average that matters, but rather the *relative*

Table 2

Nonlinearities in debt–performance interactions: prior studies

Pooled time-series cross-section instrumental least square regressions. The dependent variable is firm annual sales growth at time t , given by $(Sales_t - Sales_{t-1})/Sales_{t-1}$. *Size* is the natural logarithm of assets. *Investment* is capital expenditures over assets. *Profitability* is operating earnings over assets. *SellExpenses* is the ratio of advertising and selling expenses to total sales. In Models 1 and 2, *Leverage* is a dummy variable for firm-years with a total debt-to-asset ratio in the top three deciles of the overall sample distribution, denoted *Top3Deciles*. In Models 3 and 4, *Leverage* is computed from the predicted values from a regression of the ratio of long-term debt to assets on a proxy for asset tangibility. All variables are adjusted for their four-digit SIC industry-year means, with leverage in Models 3 and 4 further standardized by dividing it by its industry-year standard deviation (z-scored), denoted *z-Leverage*. The sample period is 1971–2000. Included firms are from industries selected at the four-digit SIC level following Clarke (1989). Only observations from industry-years containing at least ten firms are used. The estimations correct the error structure for heteroskedasticity and within-firm error clustering using the Huber-White estimator. *t*-statistics are in parentheses.

Dep. var.: <i>Sales Growth_{it}</i>	Model 1 2-Year lagged <i>Leverage</i>	Model 2 3-Year lagged <i>Leverage</i>	Model 3 2-Year lagged <i>Leverage</i>	Model 4 3-Year lagged <i>Leverage</i>
<i>Size_{it}</i>	0.0065** (7.26)	0.0062** (6.66)	0.0076** (8.43)	0.0072** (7.65)
$\sum_{k=1}^2 Profitability_{t-k}$	0.1311** (8.90)	0.1594** (8.73)	0.1225** (8.26)	0.1509** (8.28)
$\sum_{k=1}^2 Investment_{t-k}$	0.4455** (14.69)	0.4461** (13.57)	0.4333** (14.02)	0.4318** (12.75)
$\sum_{k=1}^2 SellExpenses_{t-k}$	0.0078 (0.62)	0.0103 (0.79)	0.0046 (0.38)	0.0085 (0.62)
<i>Leverage:</i>				
<i>Top3Deciles</i>	−0.0192** (−4.02)	−0.0305** (−2.87)		
<i>z-Leverage</i>			0.0140** (7.82)	0.1266** (6.48)
Adjusted- R^2	0.03	0.04	0.02	0.04
N. observations	33,113	28,893	32,578	27,443

Note: ** and * indicate statistical significance at the 1% and 5% (two-tailed) test levels, respectively.

Denote a firm's standardized predicted value of leverage by *z-Leverage*. Suppose the number of different leverage categories which debt has a distinct relation with sales growth is

- When leverage is measured as an industry-year standardized tangibility-predicted variable, the coefficient is positive.
- So the same sample can reproduce both Opler–Titman and Campello–Fluck style findings.

Economic message. The literature looked inconsistent because different empirical designs were picking up different parts of the leverage distribution.

5.3 Tables 3 and 4: locating the nonlinear effect

Read:

- Table 3 shows that the leverage categories contain many firms and observations, and that the extreme right tail is not made up only of pathological debt values.
- Table 4 shows that linearity is strongly rejected.
- With wider spline segments, the middle region becomes positive and the far-right region turns strongly negative.
- The $\pm 3\sigma$ version gives the cleanest prelude to the four-segment baseline that follows.

Economic message. This is where the paper's main empirical point first becomes sharp: moderate relative leverage helps, extreme relative leverage hurts.

5.4 Table 5: who sits in each leverage region

Read:

- The **HighLev** group has positive mean and median relative sales growth.
- The **VeryHighLev** group has slightly negative relative sales growth, but still positive median profitability.

Tables 3 and 4 below are dedicated to this identification process.

4.2.2. Nonlinearities in debt–performance interactions

Results from Models 3 and 4 in Table 2 show that if one imposes a single-line form upon the association between relative indebtedness and sales growth, then a positive coefficient obtains for debt. In this section, I look for deviations from a linear monotonic relation

Table 3

Descriptive statistics for firm debt

This table provides detailed statistics for absolute and relative-to-rival debt usage among sample firms. For each one of the various breakdowns of $z\text{-Leverage}$, the table reports the number of individual firms, the number of firm-year observations, the mean and median raw leverage ratio, and the mean and median industry-year-adjusted leverage ratio. The raw leverage ratios are presented in percentage terms and the adjusted leverage ratios are presented in terms of standard deviations from the industry-year mean of Leverage . The sample period is 1971–2000. Included firms are from industries selected at the four-digit SIC level following Clarke (1989). Only observations from industry-years containing at least ten firms are used.

Breakdown of $z\text{-Leverage}$ by range	Number of firms	Number of firm-years	Raw leverage in % terms mean [Median]	Adjusted leverage in σ 's terms mean [Median]
$-\infty < z\text{-Leverage} \leq 0$	5,257	22,556	17.573 [13.951]	−0.781 [−0.582]
$0 < z\text{-Leverage} < \infty$	4,697	20,428	23.052 [20.372]	0.683 [0.530]
$-\infty < z\text{-Leverage} \leq -1.5\sigma$	1,301	3,274	10.821 [4.753]	−2.147 [−1.958]
$-1.5\sigma < z\text{-Leverage} \leq +1.5\sigma$	6,012	35,780	20.204 [17.303]	0.047 [0.0739]
$+1.5\sigma < z\text{-Leverage} \leq +\infty$	1,253	3,940	25.348 [21.909]	1.978 [1.842]
$-\infty < z\text{-Leverage} \leq -2\sigma$	810	2,412	9.157 [2.988]	−2.659 [−2.408]
$-2\sigma < z\text{-Leverage} \leq +2\sigma$	6,267	37,393	20.146 [17.110]	0.038 [0.071]
$+2\sigma < z\text{-Leverage} < +\infty$	742	3,189	26.154 [22.330]	2.443 [2.311]
$-\infty < z\text{-Leverage} \leq -3\sigma$	388	1,121	7.100 [1.558]	−3.732 [−3.433]
$-3\sigma < z\text{-Leverage} \leq +3\sigma$	6,679	39,424	20.125 [17.005]	0.016 [0.064]
$+3\sigma < z\text{-Leverage} < +\infty$	394	2,449	28.755 [25.66]	3.499 [3.309]

Table 4

Nonlinearities in debt–performance interactions: base tests

Pooled time-series cross-section instrumental least square regressions. The dependent variable is firm annual sales growth at time t , given by $(\text{Sales}_t - \text{Sales}_{t-1})/\text{Sales}_{t-1}$. Size_t is the natural logarithm of assets. Investment is capital expenditures over assets. Profitability is operating earnings over assets. SellExpenses is the ratio of advertising and selling expenses to total sales. Leverage is computed from the predicted values from a regression of the ratio of long-term debt to assets on a proxy for asset tangibility. All variables are adjusted for their four-digit SIC industry-year means, with leverage further standardized by dividing it by its industry-year standard deviation (z -scored), denoted $z\text{-Leverage}$. The sample period is 1971–2000. Included firms are from industries selected at the four-digit SIC level following Clarke (1989). Only observations from industry-years containing at least ten firms are used. The estimations correct the error structure for heteroskedasticity and within-firm error clustering using the Huber-White estimator. t -statistics are in parentheses.

Dep. var.: Sales Growth_t	Model 1	Model 2	Model 3	Model 4	Model 5
Size_t	0.0076** (8.43)	0.0078** (8.62)	0.0079** (8.78)	0.0078** (8.72)	0.0077** (8.61)
$\sum_{k=1}^2 \text{Profitability}_{t-k}$	0.1225** (8.26)	0.1234** (8.33)	0.1227** (8.30)	0.1224** (8.29)	0.1222** (8.27)
$\sum_{k=1}^2 \text{Investment}_{t-k}$	0.4333** (14.02)	0.4423** (14.22)	0.4357** (14.06)	0.4353** (14.07)	0.4341** (14.05)
$\sum_{k=1}^2 \text{SellExpenses}_{t-k}$	0.0046 (0.38)	0.0063 (0.52)	0.0047 (0.39)	0.0043 (0.35)	0.0042 (0.35)
Leverage_{t-1}	0.0140** (7.82)				
$-\infty < z\text{-Leverage} \leq 0$		0.0066* (2.48)			
$0 < z\text{-Leverage} < \infty$		0.0222** (6.43)			
$-\infty < z\text{-Leverage} \leq -1.5\sigma$			−0.0189** (−2.92)		
$-1.5\sigma < z\text{-Leverage} \leq +1.5\sigma$			0.0200** (8.92)		
$+1.5\sigma < z\text{-Leverage} \leq +\infty$			−0.0037 (−0.30)		
$-\infty < z\text{-Leverage} \leq -2\sigma$				−0.0315** (−4.10)	
$-2\sigma < z\text{-Leverage} \leq +2\sigma$				0.0179** (9.04)	
$+2\sigma < z\text{-Leverage} < +\infty$				−0.0190 (−0.90)	
$-\infty < z\text{-Leverage} \leq -3\sigma$					−0.0389** (−5.54)
$-3\sigma < z\text{-Leverage} \leq +3\sigma$					0.0158** (8.79)
$+3\sigma < z\text{-Leverage} < +\infty$					−0.0895** (−3.83)
Partial F -tests for Leverage coefficients					
F -statistic (null of joint insignificance)		34.60**	28.28**	28.95**	34.58**
F -statistic (null of linearity)		7.74**	14.32**	18.33**	32.03**
Adjusted- R^2	0.02	0.04	0.05	0.05	0.06
N. observations	32,578	32,578	32,578	32,578	32,578

Note: ** and * indicate statistical significance at the 1% and 5% (two-tailed) test levels, respectively.

- Tangibility rises as one moves from very low toward very high leverage regions.

Economic message. The positive middle-range effect is not just a mechanical artifact: the firms that sit there are not random, and the very-high-leverage firms are not simply all distressed or unprofitable.

5.5 Tables 6, 7, 8 and Figure 1: detailed characterization and economic magnitudes

Read:

- Table 6 reports IV estimates for the four-segment spline. The sign pattern is:

$$\text{VeryLowLev} < 0, \quad \text{LowLev} > 0, \quad \text{HighLev} \gg 0, \quad \text{VeryHighLev} < 0.$$

- Table 7 shows that the same qualitative pattern remains under GMM, even after instrumenting investment and sell expenses as well.
- Table 8 translates the baseline coefficients into economic magnitudes: roughly -4.1% , $+0.7\%$, $+2.1\%$, and -16.4% for the four segments in the GMM baseline.
- Figure 1 plots the implied spline. It rises from negative values at the left tail, climbs modestly around the center, peaks in the high-leverage region, and then collapses sharply in the very-high-leverage region.

Table 5

Summary statistics for (within-industry-year) debt categories

This table reports summary statistics for the main variables used in estimations (after industry-year adjustments) separately for each of the four debt categories considered in the baseline four-segment spline model. *Sales Growth* is the firm annual sales growth. *Size* is the natural logarithm of assets. *Investment* is capital expenditures over assets. *Profitability* is operating earnings over assets. *SellExpenses* is the ratio of advertising and selling expenses to total sales. *Tangibility* is computed as a weighted sum of cash holdings, accounts receivables, inventories, and net fixed capital, divided by total assets. *Leverage* is computed from the predicted values from a regression of the ratio of long-term debt to assets on a proxy for asset tangibility. All variables are adjusted for their four-digit SIC industry-year means. The debt categories *VeryLowLev* and *VeryHighLev* group firms with debt-to-asset ratios below -3σ and above $+3\sigma$ from the industry-year average, respectively. *LowLev* and *HighLev* group firms whose leverage are in the -3σ to 0 range and in the 0 to $+3\sigma$ range of the industry-year standardized leverage distribution, respectively; 0 is the industry-year average. Observations are from industry-years with at least ten firms.

Industry-year adj. variables	Debt categories (ranked within industry-years)			
	<i>VeryLowLev</i> (# Firms = 388) (# Obs. = 1,121)	<i>LowLev</i> (# Firms = 4,855) (# Obs. = 21,445)	<i>HighLev</i> (# Firms = 4,333) (# Obs. = 17,799)	<i>VeryHighLev</i> (# Firms = 394) (# Obs. = 2,449)
<i>Size</i>				
Mean	-1.2080	0.3219	-0.2683	0.0912
Median	-1.1452	0.2635	-0.2710	-0.0119
Std. Dev.	2.4500	2.0682	1.8441	2.0561
Inteq. range	3.4712	2.8204	2.3892	2.6706
<i>Sales Growth</i>				
Mean	-0.0921	-0.0200	0.0185	-0.0062
Median	-0.0596	-0.0128	0.0051	-0.0016
Std. Dev.	0.2473	0.2412	0.2545	0.2160
Inteq. range	0.2425	0.2068	0.2246	0.1785
<i>Profitability</i>				
Mean	-0.1507	-0.0077	0.0041	-0.0025
Median	-0.0459	0.0106	0.0312	0.0229
Std. Dev.	0.4556	0.3312	0.4750	0.4752
Inteq. range	0.1830	0.1136	0.1100	0.1236
<i>Investment</i>				
Mean	-0.0652	-0.0051	0.0052	0.0043
Median	-0.0940	-0.0125	-0.0033	-0.0038
Std. Dev.	0.0652	0.0650	0.0700	0.0962
Inteq. range	0.0821	0.0523	0.0588	0.0670
<i>SellExpenses</i>				
Mean	0.0201	-0.0125	0.0123	-0.0290
Median	-0.0422	-0.0436	-0.0268	-0.0463
Std. Dev.	0.4475	0.3012	0.2978	0.3451
Inteq. range	0.1710	0.1742	0.1853	0.1359
<i>Tangibility</i>				
Mean	-0.1352	-0.0705	0.0632	0.0998
Median	-0.1260	-0.0468	0.0430	0.0859
Std. Dev.	0.0777	0.0712	0.0642	0.0880
Inteq. range	0.0622	0.0795	0.0663	0.0952

Table 6

Nonlinearities in debt performance interactions: detailed characterization (IV estimations)

Pooled time-series cross-section regressions. The dependent variable is firm annual sales growth at time t , given by $(\Delta \text{Sales}_t - \text{Sales}_{t-1})/\text{Sales}_{t-1}$. *Size* is the natural logarithm of assets. *Investment* is capital expenditures over assets. *Profitability* is operating earnings over assets. *SellExpenses* is the ratio of advertising and selling expenses to total sales. *Leverage* is computed from the predicted values from a regression of the ratio of long-term debt to assets on a proxy for asset tangibility. All variables are adjusted for their four-digit SIC industry-year means, with leverage further standardized by dividing it by its industry-year standard deviation (z -scored). *zLeverage*, *zVeryLowLev* and *zVeryHighLev* denote leverage levels below -3σ and above $+3\sigma$ from the industry-year average, respectively (i.e., $zLeverage > 3$ or $zLeverage < -3$). *LowLev* and *HighLev* denote leverage levels in the -3σ to 0 range and in the 0 to $+3\sigma$ range of the industry-year standardized leverage distribution, respectively; 0 is the industry-year average. The sample period is 1971–2000. The estimations correct the error structure for heteroskedasticity and within-firm error clustering using the Huber-White estimator. t -statistics are in parentheses.

Dep. var.: Sales Growth	Model 1 Baseline model	Model 2 3-Year lag Lev	Model 3 # Firms >40	Model 4 -1 ≤ Skewness ≤ 1	Model 5 Total debt	Model 6 Firm effects
<i>Size</i>	0.0079** (8.73)	0.0073** (7.80)	0.0077** (8.83)	0.0065** (5.88)	0.0078** (8.72)	0.0033** (4.68)
$\Sigma_{i=1}^3 \text{Profitability}_{t-i}$	0.1250** (8.32)	0.0884** (5.51)	0.0794** (4.85)	0.2136** (7.14)	0.1254** (8.35)	0.1486** (8.81)
$\Sigma_{i=1}^3 \text{Investment}_{t-i}$	0.4411** (14.19)	0.4339** (11.60)	0.4442** (11.91)	0.3163** (13.10)	0.4413** (14.20)	0.4333** (13.98)
$\Sigma_{i=1}^3 \text{SellExpenses}_{t-i}$	0.0057 (0.47)	0.0008 (0.08)	-0.0253 (-1.37)	0.0213 (1.41)	0.0063 (0.52)	0.0037 (0.30)
<i>Leverage</i> , z : (spline segments)						
<i>VeryLowLev</i>	-0.0520** (-5.55)	-0.0372** (-3.01)	-0.0520** (-5.50)	-0.0460** (-5.54)	-0.0520** (-5.55)	-0.0480** (-5.43)
<i>LowLev</i>	0.0067* (2.22)	0.0057* (2.91)	0.0067* (1.77)	0.0067* (2.33)	0.0067** (2.79)	0.0067** (2.79)
<i>HighLev</i>	0.0225** (6.75)	0.0225** (5.31)	0.0225** (6.04)	0.0225** (5.33)	0.0225** (6.74)	0.0225** (6.80)
<i>VeryHighLev</i>	-0.0986** (-4.27)	-0.1089** (-5.03)	-0.0952** (-4.50)	-0.0986** (-4.50)	-0.0986** (-4.50)	-0.1048** (-4.50)
Adjusted R^2	0.05	0.04	0.05	0.05	0.04	0.04
Partial F -statistic (linearity)	26.25** (32.578)	18.83** (27.483)	27.66** (35.970)	18.30** (23.598)	20.20** (32.582)	27.82** (32.578)
N observations	32,578	32,578	32,578	32,578	32,578	32,578

Note: ** and * indicate statistical significance at the 1% and 5% (two-tailed) test levels, respectively.

Economic message. The debt–performance relation is not just statistically nonlinear; it is economically large and highly asymmetric across leverage regions.

Dep. var.: Sales Growth	Model 1 Baseline model		Model 2 3-Year lag Lev		Model 3 # Firms >40		Model 4 -1 ≤ Skewness ≤ 1		Model 5 Total debt		Model 6 Firm effects	
	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates	IV estimates
<i>Size</i>	0.0079** (8.73)	0.0073** (7.80)	0.0077** (8.83)	0.0065** (5.88)	0.0078** (8.72)	0.0033** (4.68)	0.0079** (8.73)	0.0073** (7.80)	0.0077** (8.83)	0.0065** (5.88)	0.0078** (8.72)	0.0033** (4.68)
$\Sigma_{i=1}^3 \text{Profitability}_{t-i}$	0.1250** (8.32)	0.0884** (5.51)	0.0794** (4.85)	0.2136** (7.14)	0.1254** (8.35)	0.1486** (8.81)	0.1250** (8.32)	0.0884** (5.51)	0.0794** (4.85)	0.2136** (7.14)	0.1254** (8.35)	0.1486** (8.81)
$\Sigma_{i=1}^3 \text{Investment}_{t-i}$	0.4411** (14.19)	0.4339** (11.60)	0.4442** (11.91)	0.3163** (13.10)	0.4413** (14.20)	0.4333** (13.98)	0.4411** (14.19)	0.4339** (11.60)	0.4442** (11.91)	0.3163** (13.10)	0.4413** (14.20)	0.4333** (13.98)
$\Sigma_{i=1}^3 \text{SellExpenses}_{t-i}$	0.0057 (0.47)	0.0008 (0.08)	-0.0253 (-1.37)	0.0213 (1.41)	0.0063 (0.52)	0.0037 (0.30)	0.0057 (0.47)	0.0008 (0.08)	-0.0253 (-1.37)	0.0213 (1.41)	0.0063 (0.52)	0.0037 (0.30)
<i>Leverage</i> , z : (spline segments)												
<i>VeryLowLev</i>	-0.0520** (-5.55)	-0.0372** (-3.01)	-0.0520** (-5.50)	-0.0460** (-5.54)	-0.0520** (-5.55)	-0.0480** (-5.43)	-0.0520** (-5.55)	-0.0372** (-3.01)	-0.0520** (-5.50)	-0.0460** (-5.54)	-0.0520** (-5.55)	-0.0480** (-5.43)
<i>LowLev</i>	0.0067* (2.22)	0.0057* (2.91)	0.0067* (1.77)	0.0067* (2.33)	0.0067** (2.79)	0.0067** (2.79)	0.0067* (2.22)	0.0057* (2.91)	0.0067* (1.77)	0.0067* (2.33)	0.0067** (2.79)	0.0067** (2.79)
<i>HighLev</i>	0.0225** (6.75)	0.0225** (5.31)	0.0225** (6.04)	0.0225** (5.33)	0.0225** (6.74)	0.0225** (6.80)	0.0225** (6.75)	0.0225** (5.31)	0.0225** (6.04)	0.0225** (5.33)	0.0225** (6.74)	0.0225** (6.80)
<i>VeryHighLev</i>	-0.0986** (-4.27)	-0.1089** (-5.03)	-0.0952** (-4.50)	-0.0986** (-4.50)	-0.0986** (-4.50)	-0.1048** (-4.50)	-0.0986** (-4.27)	-0.1089** (-5.03)	-0.0952** (-4.50)	-0.0986** (-4.50)	-0.0986** (-4.50)	-0.1048** (-4.50)
Adjusted R^2	0.05	0.04	0.05	0.05	0.04	0.04	0.05	0.04	0.05	0.05	0.04	0.04
Partial F -statistic (linearity)	26.25** (32.578)	18.83** (27.483)	27.66** (35.970)	18.30** (23.598)	20.20** (32.582)	27.82** (32.578)	26.25** (32.578)	18.83** (27.483)	27.66** (35.970)	18.30** (23.598)	20.20** (32.582)	27.82** (32.578)
N observations	32,578	32,578	32,578	32,578	32,578	32,578	32,578	32,578	32,578	32,578	32,578	32,578

Note: ** and * indicate statistical significance at the 1% and 5% (two-tailed) test levels, respectively.

positive strategic effect on product market performance. In contrast, similar leverage increases at lower levels of relative-to-industry indebtedness lead to either sales growth underperformance (about -4.1% in the *VeryLowLev* range) or to economically small and statistically marginal gains (0.7% in the *LowLev* range). Finally, the coefficient returned

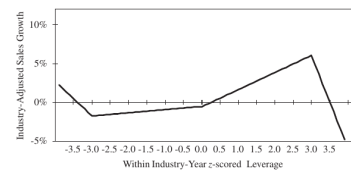
Table 8

Nonlinearities in debt performance interactions: economic magnitudes

The table synthesizes the economic and statistical significance of the impact of debt taking on product market performance. Each of the cells represents the response of a firm's relative-to-industry sales growth (in percentage terms) to a one-standard deviation change in the firm's tangibility-predicted leverage ratio across different debt categories. The four debt categories of the baseline specification are derived from industry-year normalized debt-to-asset ratios. Model 1 reports estimates obtained from the baseline four-segment spline IV estimation of Table 6. Model 2 reports similar results from the GMM estimation of Table 7. p -values are in square brackets.

Leverage spline segments	Model 1 IV estimates		Model 2 GMM estimates	
	% sales growth response to one-Std. Dev. change in debt	% sales growth response to one-Std. Dev. change in debt	% sales growth response to one-Std. Dev. change in debt	% sales growth response to one-Std. Dev. change in debt
<i>VeryLowLev</i>	-5.20 [0.00]	-4.11 [0.00]	-5.20 [0.00]	-4.11 [0.00]
<i>LowLev</i>	0.69 [0.03]	0.69 [0.03]	0.69 [0.03]	0.69 [0.03]
<i>HighLev</i>	2.28 [0.00]	2.28 [0.00]	2.28 [0.00]	2.28 [0.00]
<i>VeryHighLev</i>	-16.39 [0.00]	-16.39 [0.00]	-16.39 [0.00]	-16.39 [0.00]

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s growth response to debt at varying levels of leverage (spline function). The figure depicts industry-year adjusted sales response to the lagged industry-year normalized, it four different segments of the normalized leverage ($-\infty, -3\sigma$; $-3\sigma, 0$; $+3\sigma, 0$; $+3\sigma, \infty$) GMM-estimated coefficients from Model 1 of Table 5. The initial point of the function listed sales growth for the unleveraged (zero-debt) firms in the sample, 2.4%.

5.6 Figure 2 and Table 9: concentration and firm leadership

Read:

- Figure 2 shows that the leverage curve is much steeper in concentrated industries than in competitive industries.
- In concentrated industries, the gain from moving into the **HighLev** region is larger, and the loss from entering **VeryHighLev** is much more severe.
- Table 9 shows that in concentrated industries, ordinary firms with moderately high leverage gain relative sales, but leaders do not: the **HighLev** $\times$ **Leader** interaction is negative.

- Conversely, low-leverage leaders in concentrated industries benefit from higher leverage: the $\text{LowLev} \times \text{Leader}$ interaction is positive.
- In competitive industries, most leader interactions are small and statistically weak.

Economic message. Strategic debt matters most where industry structure makes competitive responses meaningful. In concentrated markets, debt is a sharper strategic instrument—but not for already-dominant leaders with too much leverage.

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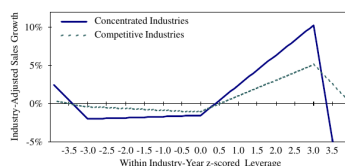


Fig. 1. growth response to debt at varying levels of leverage: concentrated versus competitive industries. The estimated expected industry-year adjusted sales response to the lagged instrumented leverage at four different segments of the normalized leverage: $(-\infty, -3]$, $(-3, 0]$, $(0, 3]$, and $(3, \infty)$. The slopes are estimated via GMM (separately) over samples of concentrated and competitive manufacturing industries.

6 Short summary

- The paper argues that debt can both help and hurt product-market performance.
- The theory delivers a nonmonotonic relation because moderate debt can support aggressive conduct, but excessive debt cannot.
- The empirical design compares firms to rivals within the same industry-year and instruments leverage with tangibility.
- Relative sales growth is mildly positive for low leverage, strongly positive for moderately high leverage, and sharply negative for very high leverage.
- The pattern survives alternative lags, alternative leverage measures, skewness restrictions, firm fixed effects, and GMM.
- The effect is much stronger in concentrated industries.
- In concentrated markets, moderately high leverage helps ordinary firms but not leaders; leaders benefit more from leverage only when they begin from conservative debt positions.

7 Conclusion

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Table 9

Nonlinearities in debt performance interactions: industry concentration and firm leadership
Pooled time-series cross-section regressions. The dependent variable is firm annual sales growth at time t , given by $\Delta \ln(\text{Sales}_{i,t})$. Size is the natural logarithm of assets. Investment is capital expenditures over assets. Profitability is operating earnings over assets. SelfExpense is the ratio of advertising and selling expenses to total sales. Leverage is computed from the predicted values from a regression of the ratio of long-term debt to assets on a proxy for asset tangibility. All variables are adjusted for their first-stage SIC industry-year means, with leverage further standardized by dividing it by its industry-year standard deviation (z -scored). $z\text{-Leverage}$, $z\text{-Profitability}$, and $z\text{-SelfExpense}$ denote leverage levels below -3σ and above $+3\sigma$ from the industry-year average, respectively (i.e., $z\text{-Leverage} > 3\sigma$). $LowLev$ and $HighLev$ denote leverage levels in the -3σ to 0 range and in the 0 to $+3\sigma$ range of the industry-year standardized leverage distribution, respectively. 0 is the industry-year average. Observations are from industry-years with at least ten firms. Concentrated industries are those with Herfindahl–Hirschman (HH) index ≥ 1000 , while competitive industries are those with HH index < 1000 . A market leader (Leader) is a firm with total sales accounting for more than 15% of the total industry-year sales. The sample period is 1980–1994 and sampling is restricted to manufacturing industries (SIC 2000–3999) due to limitations in the Bureau of Census coverage on industry concentration. The estimations correct the error structure for heteroskedasticity and within-firm error clustering using the Huber–White estimator. t -statistics are in parentheses. GMM estimations display diagnostic statistics for instrument overidentification restrictions (p -values for Hansen’s J -statistics reported) and instrument first-stage partial F -statistics (lowest F -statistics reported).

Dep. var.: Sales Growth	Model 1 All industries	Model 2 Concentrated inds. HH-index ≥ 1000 (A)	Model 3 Competitive inds. HH-index < 1000 (B)	Diff. p -value (A) – (B)
<i>Panel A: IV regressions</i>				
Size _{$t-1$}	0.0025* (0.37)	0.0071 (1.74)	0.0031 (1.21)	0.41
$\Sigma_{i=1}^4 \text{Profitability}_{i,t-1}$	0.151** (0.01)	0.198** (2.58)	0.1396** (1.60)	0.44
$\Sigma_{i=1}^4 \text{Investment}_{i,t-1}$	0.4256** (0.27)	0.4404** (2.36)	0.3158** (1.70)	0.54
$\Sigma_{i=1}^4 \text{SelfExpense}_{i,t-1}$	0.0054 (0.27)	0.0001 (0.53)	–0.0091 (–0.39)	0.51
$VeryLowLev_{i,t-1}$	–0.109** (–2.80)	–0.048 (–1.44)	–0.2684 (–1.43)	0.23
$LowLev_{i,t-1}$	0.0020 (0.53)	–0.010 (–1.46)	0.0029 (0.70)	0.11
$HighLev_{i,t-1}$	0.026** (4.44)	0.042** (4.35)	0.0172* (2.27)	0.01
$VeryHighLev_{i,t-1}$	–0.4483 (–1.26)	–0.822** (–15.47)	–0.0958 (–1.29)	0.00
$VeryLowLev \times Leader_{i,t-1}$	0.0161 (1.92)	0.0730** (4.97)	0.0047 (0.44)	0.00
$LowLev \times Leader_{i,t-1}$	0.0102 (0.80)	0.0803** (4.44)	0.0027 (0.11)	0.00
$HighLev \times Leader_{i,t-1}$	–0.0075 (–0.40)	–0.0801** (–2.36)	0.0027 (0.11)	0.02
Adjusted R^2	0.04	0.97	0.02	
N. observations	8,923	1,559	5,466	
<i>Panel B: GMM regressions</i>				
Size _{$t-1$}	0.0025 (1.63)	0.0051 (1.81)	0.0011 (1.39)	0.18

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Table 9 (continued)

Dep. var.: Sales Growth	Model 1 All industries	Model 2 Concentrated inds. HH-index ≥ 1000 (A)	Model 3 Competitive inds. HH-index < 1000 (B)	Diff. p -value (A) – (B)
$\Sigma_{i=1}^4 \text{Profitability}_{i,t-1}$	0.2036* (2.23)	0.2415* (2.07)	0.2822** (2.80)	0.79
$\Sigma_{i=1}^4 \text{Investment}_{i,t-1}$	0.5354* (1.90)	0.6140 (1.83)	0.4099 (0.312)	0.63
$\Sigma_{i=1}^4 \text{SelfExpense}_{i,t-1}$	0.2623* (2.00)	0.2222** (2.91)	0.189 (1.89)	0.62
$VeryLowLev_{i,t-1}$	–0.100** (–2.25)	–0.069 (–1.55)	–0.186** (–2.43)	0.56
$LowLev_{i,t-1}$	–0.0025 (–0.18)	–0.013 (–0.97)	–0.0038 (–0.24)	0.65
$HighLev_{i,t-1}$	0.027** (2.25)	0.0726** (4.64)	0.0217 (1.56)	0.02
$VeryHighLev_{i,t-1}$	–0.465** (–3.26)	–0.761** (–8.60)	–0.0217 (1.05)	0.00
$VeryLowLev \times Leader_{i,t-1}$	0.0118 (1.74)	0.0355* (2.27)	–0.0084 (–0.76)	0.02
$LowLev \times Leader_{i,t-1}$	0.0170* (2.33)	0.0771** (4.71)	0.0067 (0.55)	0.00
$HighLev \times Leader_{i,t-1}$	–0.0029 (–0.15)	–0.0350** (–2.33)	0.0070 (0.35)	0.06
Hansen’s J -statistic p -value	0.67	0.26	0.08	
Lowest first-stage F -statistic	92.68**	24.14**	44.17**	
N. observations	7,083	1,233	4,331	

Note: ** and * indicate statistical significance at the 1% and 5% (two-tailed) test levels, respectively.

7.1 Core conclusion (what the paper establishes)

The paper establishes that the relationship between debt and product-market performance is **non-linear and nonmonotonic**. Moderate debt can raise relative sales performance, but very high debt is associated with sharp product-market underperformance.

7.2 Economic intuition

Debt can act like a commitment device because it changes the firm's incentives and the beliefs of rivals. But commitment is not unlimited. Once debt becomes too extreme, either contracts are renegotiated away from inefficient aggression or the burden of debt itself damages the firm's competitive position. The result is an inverted-U type response in the strategic value of leverage.

7.3 Takeaway

The main lesson is not that "debt is good" or "debt is bad." It is that **where the firm sits in the leverage distribution relative to rivals** matters enormously. Moderate leverage can be a strategic asset; excessive leverage becomes a competitive liability.